

Algebraic Genomics: Contents and Synthesis of the Monograph

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Code and data: <https://github.com/Ruqing1963/algebraic-genomics>

The monograph collects the *Bio* papers of the applied strand of the series, together with one co-authored paper. Series numbers are those written in the files; for Bio 1–7 no number is written, and the numbers XIV–XX follow from Bio 8 = XXI by position. Bio 15, 16 and 18 were specified and then dropped, so XXVIII, XXIX and XXXI are unused. Every chapter comes with a verification script that reproduces its numbers.

The one-paragraph version

A genome's k -mer data determine it only up to a *fibre*: the set of sequences with the same data. For one strand the fibre is the set of Eulerian circuits of a de Bruijn graph. Its size is the BEST determinant, which factors into a local part and a global part, the order of the sandpile group, and the sandpile group splits along the repeats (Part I). For two strands the fibre is a union of such sets over the points of a lattice in a box. Its size is a lattice sum of determinants. We compute it exactly for $\phi X174$ at $k = 10$, $\#DS = 31\,925\,753\,246\,212$, and prove that no closed formula exists: the problem is $\#P$ -hard (Part II). The fibre is traversed by explicit moves. These are chips on one strand, and transpositions and inversion flips on two, with a Tate-cohomology dichotomy for the lattice the flips span. The same fibres are exactly what local sequence encoders cannot see (Part III). Where the problem is tractable it is so for a reason that can be named: bounded frontier or snarl width, repeat-only compaction, or linear structure. Those reasons give linear-time algorithms on real genomes (Part IV).

Contents

ch.	paper	title and core theorem
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Part I. Classical foundations: the BEST theorem and the sandpile group

ch.	paper	title and core theorem
1	Bio 1 (XIV)	<i>Assembly ambiguity is not determined by branch statistics: critical groups of de Bruijn graphs.</i> BEST = local factor $\prod(d^+ - 1)! \times$ global factor $ \mathcal{K}(G_k) $. Equal degree data can give 4 versus 256 reconstructions. A repeat of length L and multiplicity r contributes $(\mathbb{Z}/r)^{L-k}$, and interleaving adds a $\mathbb{Z}/2$ that nested or disjoint repeats do not.
2	Bio 2 (XV)	<i>The sandpile group as a coordinate system on genome reconstructions.</i> $\mathcal{K}(G)$ acts simply transitively on arborescences by rotor-routing. One chip is an explicit rearrangement between two candidate genomes. $\phi X174$ at $k = 12$ has $\mathcal{K} = \mathbb{Z}/2$ and exactly two reconstructions, exchanged by one chip.
3	Bio 3 (XVI)	<i>The sandpile group of a de Bruijn graph splits along its repeats.</i> Bundle-chain lemma: an r -bundle chain of length s splits off $(\mathbb{Z}/r)^{s-1}$. So $\mathcal{K}(G_k) \cong \bigoplus_c (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk})$, a length-driven part plus the group of the repeat-adjacency skeleton, which no k removes. For two-copy repeats it is $\text{coker}(A + I)$ of Bouchet's interlace matrix.
4	Bio 4 (XVII)	<i>Multi-copy repeats: where the two-copy theory stops.</i> For two families, $\mathcal{K} = \mathbb{Z}/c$ with c the number of transitions. For three-copy repeats no pairwise presentation exists: equal pairwise patterns give $\mathbb{Z}/3$ and $\mathbb{Z}/4$. A census of realized groups.

Part II. The double-stranded frontier and its complexity

5	Bio 6 (XIX)	<i>Double strands without a new lemma: the reverse-complement double cover.</i> D_k , built from the $(k+1)$ -mers of S and $\bar{S}$, is balanced, so all single-strand lemmas apply verbatim. Multiplicities add across strands. D_k is connected iff S has an inverted repeat of length $\geq k$. The involution ρ reverses arcs.
6	Bio 7 (XX)	<i>The reverse-complement quotient is a signed graph.</i> The quotient D_k/ρ is a Zaslavsky signed graph, unbalanced iff there is an inverted repeat, and the Reiner–Tseng sequence holds. But $P\Delta P = \Delta^\top$: ρ is an anti-automorphism, and the directed sandpile group lies in no covering sequence. What ρ gives is a symmetric linking form.
7	Bio 8 (XXI)	<i>Double-stranded genome assembly as a lattice sum, and the inversion-flip group.</i> $\#\text{DS}(S, k) = \sum_{f \in \Lambda^-, f \leq m} N_{\text{seq}}((m+f)/2)$, with the mandatory division by $\prod c(a)!$. $\text{rank } \Lambda^- = (A - V + 2 - \#\text{fix}_A - \#\text{fix}_V)/2$ by the Hopf trace formula. Inversion flips preserve the double-stranded spectrum. $\phi X174$: $\#\text{DS} = 10$ at $k = 12$ and 864 at $k = 11$, and the flip graph is connected there.
8	Bio 12 (XXV)	<i>The double-stranded lattice sum by frontier elimination.</i> The cost is set by the frontier width, not the rank. $\phi X174$ at $k = 10$ has 2 481 687 900 box points, with 50 758 states and 26 open vertices; 909 103 210 of them have connected support. Connectivity is not a side condition but the non-vanishing of $t(G_c)$.

ch.	paper	title and core theorem
9	Bio 13 (XXVI)	<i>The double-stranded lattice sum as a fermionic partition function.</i> $t(G_c)$ is a Berezin integral with one even factor per arc. Fixing a root on the half-box and using $c \mapsto m - c$ halves the box; integrality follows. The fermionic and arborescence frontiers are exact but too wide. Compiled enumeration with forced-arc contraction gives $\#DS(\phi X174, 10) = 31\,925\,753\,246\,212$.
10	Bio 19 (XXXII)	<i>Why there is no closed formula for $\#DS$.</i> From any graph G of maximum degree 4 we build a genome with $\#DS = 2\varepsilon(G)$ using palindromic vertices. So $\#DS$ and the box-point count are $\#P$ -hard, and no polynomial or “meshless” formula exists unless $FP = \#P$. Odd k is open.

Part III. Spectral fibres and rearrangements

11	Bio 17 (XXX)	<i>Inversions at inverted repeats and the double-stranded fibre.</i> Flips (Bio 8) together with transpositions and reverse complement generate $DS(S, k)$ in every case closed. This is stated as a conjecture, and proved on the class of Bio 19 by Kotzig’s theorem. The Tate dichotomy: $[\Lambda^- : (1 - \rho)Z_1] = 2$ iff there is no palindromic vertex and no self-complementary content, and the flip lattice is always Λ^- .
12	Chen & Chen	<i>What a local encoder cannot see: spectral fibres of the human proteome.</i> Encoders local of order k are constant on fibres, and fibre sizes are exact BEST counts. Across the reviewed human proteome the median κ is 5. Separation of the proteome is not resolution. The unresolved order lives in segmental-duplication families and KZFPs. Explicit PseAAC and CNN collisions are exhibited. ¹
13	Bio 9 (XXII)	<i>Polyploid phasing as a p-fold decomposition of a balanced flow.</i> Labelled phasings form a lattice sum over ordered decompositions $m = c_1 + \cdots + c_p$. Sym_p gives a homogeneous space of alleles, not a monodromy.
14	Bio 10 (XXIII)	<i>The pan-genome sheaf sees nothing.</i> The natural sheaf is a direct sum. The information lies in the recombination group $\text{Rec} = Z_1(G) / \sum_i Z_1(G_i)$, which is the H_1 of the nerve and can have torsion ($\mathbb{Z}/2$ for $\mathbb{RP}^2$ -patterned strains).

Part IV. Algorithmic implementations on genomes

15	Bio 5 (XVIII)	<i>The bundle-chain decomposition of a bacterial genome: <i>E. coli</i> K-12.</i> A vertex survives unitig contraction iff its k -mer is repeated. At $k = 150$, 4 641 652 vertices reduce to 23 933 and then to 115 in 13 seconds. The length-driven summand is 10^{10439} and the irreducible one $10^{49.7}$. The ribosomal operons split by strand.
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¹Joint work with Zhengyi Chen (Guangxi Key Laboratory of Drug Discovery and Optimization, School of Pharmacy, Guilin Medical University), published separately; included here with the co-author’s agreement.

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16	Bio 14 (XXVII)	<i>The BEST determinant along the snarl tree.</i> Snarl elimination is a Schur complement. The boundary data form a Laplacian, and $\det L_{II}$ counts boundary-rooted forests. Snarl order runs in linear time: front 8 and $\approx 10 \mu\text{s}$ per vertex up to 39 714 vertices.
17	Bio 11 (XXIV)	<i>The isoform decomposition polytope of a splicing graph.</i> $\dim \Theta(w) = \#\text{paths} - (E - V + 2)$. Deconvolution is unique for one alternative event and fails for two. Vertices of Θ need not be integral. A BEST identity holds for the integer points.

How the chapters interlock

One strand (Part I). BEST splits the count into local and global parts. The global part is a group. Bio 1 shows what the group records, Bio 2 turns it into coordinates (chips are moves), Bio 3 splits it along the repeats, and Bio 4 finds where pairwise data stop. Part IV uses Bio 3’s splitting theorem (Bio 5) and Bio 1’s determinant (Bio 14).

Two strands (Part II). The double cover (Bio 6) keeps every lemma, but its involution is an anti-automorphism (Bio 7). So there is no quotient sandpile group to count with, and the count must be assembled point by point (Bio 8). Bio 12 organizes the points by a frontier. Bio 13 identifies the determinant as a fermionic integral and evaluates ϕ_{X174} at $k = 10$. Bio 19 explains why nothing better is possible in general.

Moves (Parts I and III). Every fibre has explicit moves. On one strand there are chips (Bio 2) and transpositions (Ukkonen–Pevzner). On two strands there are flips (Bio 8), which generate a lattice that Bio 17 computes: always Λ^- , with a $\mathbb{Z}/2$ gap for cycle flips governed by palindromes. Bio 17’s generation conjecture is proved on Bio 19’s hard instances.

Fibres as blind spots (Part III). The fibre that bounds assembly also bounds representation. A local encoder cannot separate a sequence from its fibre-mates (Chen & Chen). Phasing and pan-genomes are fibres of pooled spectra: of p summands (Bio 9) and of strain walks (Bio 10).

Tractability (Part IV). Hardness is worst-case. Real genomes are tractable because their width is small. That is the width of the repeat skeleton (Bio 5), of the snarl tree (Bio 14), of the frontier (Bio 12), and of splicing graphs as DAGs (Bio 11).

Open problems carried by the monograph

1. The double-stranded Ukkonen–Pevzner conjecture (Bio 17) in general.
2. $\#P$ -hardness of $\#DS$ for odd k , where the quotient is a signed graph (Bio 7, Bio 19).
3. An FPRAS for $\#DS$; for undirected Eulerian circuits this is open as well.
4. The parity of flip distances between same-spectrum reconstructions (Bio 8, Remark on transpositions).

5. Interlacement formulas for r -in r -out skeleta with $r \geq 3$ (Bio 4).

Dropped

Bio 15 (Massey products), Bio 16 (recombination sheaf) and Bio 18 (chromatin gauge theory) were specified and not written. Their premises were analogies without a mathematical object: products into H^2 of a graph, which is zero; a reformulation that leaves the complexity of phasing unchanged; and a lattice gauge theory with no derivable observable.